

MARAN A

AI Engineer

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PROFESSIONAL SUMMARY

AI Engineer working across machine learning, computer vision, NLP, generative AI, document intelligence, RAG, and intelligent automation. Experienced in building AI-driven systems and production-oriented applications using Python, deep learning, LLMs, OCR, Document AI, FastAPI, cloud services, and serverless architectures. Interested in turning emerging AI technologies into useful systems for real-world problems.

EXPERIENCE

Tata Consultancy Services

Jan 2025 — Present

AI Engineer

- Design, develop, and deploy AI-driven solutions using Python, machine learning, deep learning, computer vision, NLP, LLMs, and Document AI.
- Build document intelligence workflows using Google Document AI and Gemini for classification, extraction, validation, and workflow automation.
- Develop retrieval-augmented generation applications using LLMs, vector databases, and FastAPI.
- Work on machine-learning-based resource allocation for airport lounge management.
- Develop AWS Lambda and Python serverless ETL workflows for Excel-based partner and user data processing.
- Build OCR and information extraction pipelines using OCR, NER, computer vision, and small language models.
- Develop real-time computer vision solutions using YOLO and TensorFlow Object Detection.
- Develop REST APIs using FastAPI and collaborate with architects, developers, QA teams, and stakeholders.
- Perform model evaluation, optimization, testing, deployment, and production support.

Tata Consultancy Services

Aug 2023 — Dec 2024

Systems Engineer

- Monitored enterprise infrastructure and critical applications in production environments.
- Managed production incidents, system health monitoring, troubleshooting, escalation, and resolution workflows.
- Worked with Control-M, WhatsUp Gold, Netreo, and Active Directory.
- Participated in Major Incident Management and root-cause-analysis activities.
- Built automation using Python, LLMs, and RAG to improve reporting, documentation, and operational workflows.
- Collaborated across infrastructure, application, and business teams for production support and change management.
- Supported deployment validation and continuous improvement initiatives.

Continental Autonomous Mobility

Sep 2022 — Jul 2023

Machine Learning Engineer Intern

- Developed LSTM-based vehicle trajectory prediction models using spatial-temporal attention.
- Explored federated learning using PyTorch and Flower.
- Worked with NGSIM traffic data for vehicle trajectory prediction and traffic scenario analysis.
- Performed model training, tuning, validation, and experimentation.
- Collaborated on machine learning research and technical documentation.

EDUCATION

VIT Vellore

2018 — 2023

Integrated M.Tech in Software Engineering · CGPA: 9.18 / 10

CERTIFICATION

AWS Certified Cloud Practitioner

TECHNICAL SKILLS

Programming	Python
Machine Learning	Machine Learning, Deep Learning, Computer Vision, NLP
Generative AI	LLMs, RAG, AI Agents, Document AI
Frameworks	FastAPI, PyTorch, TensorFlow, Hugging Face Transformers, LangChain, LangGraph, OpenCV, spaCy
Cloud / Infra	AWS Lambda, Amazon S3, Google Cloud, Google Cloud Storage
Databases	MySQL, SQLite, Redis, Vector Databases
Document / AI	OCR, PaddleOCR, Google Document AI, NER, Information Extraction

SELECTED AREAS OF INTEREST

Generative AI · AI Agents · Document Intelligence · Machine Learning Systems · Quantum-Inspired Optimization · Liquid Neural Networks · Topological Data Analysis · Mathematics